

Poking Holes in the Hilbert Sphere

A Working Exploration of Quantum Spatial Mapping

Drafted 6 October 2026 from a voice session about this site. The title is the one that landed in that session. Marked as a conversation product, not a claim from a physics department.

Status. Established quantum mechanics is labeled as such. The spatial-mapping proposal is labeled as a proposal. The planet-in-the-hole picture is labeled as a picture. None of the three is evidence for the others.

The hook, said badly on purpose

Someone read the front page and said the useful posture was to poke holes in the idea. That line landed harder than it should have, because there is already a worse version of the same sentence:

How do you lose a planet in the side of a Hilbert sphere? Stretch the fabric. The farther you stretch it, the bigger the holes get. Then a planet falls out. That is a terrible explanation. It is also the one that made the room laugh, so it stays at the top, where a visitor can see the tone before the notation.

The laugh is not the theory. The laugh is a warning label. If a sentence only works after you have been in the room, it does not belong on the site as an argument. It can belong as the door.

Three things that are not the same object

The phrase “Hilbert sphere” already mixes two pictures. That mix is the first hole, and it is free.

Object	What it actually is	What it is not
Hilbert space	The vector space of states. For one qubit it is two complex dimensions. For a field, or for many particles, the dimension is enormous. Inner product gives probabilities.	Not a balloon. Not the sky. Not a container that planets sit inside.
Bloch sphere	A real 2-sphere that draws the pure states of one qubit. North pole is $ 0\rangle$, south pole is $ 1\rangle$, latitude and longitude are the relative phase and mix.	Not the Hilbert space itself. A drawing of one small Hilbert space, after throwing away a global phase.
Physical space	The stage: positions, distances, a metric $g_{\mu\nu}$ if you are doing relativity. Planets live here.	Not the Bloch sphere, and not the Hilbert space, unless a theory explicitly identifies them. This site has not.

So the spoken line “a hole in the side of a Hilbert sphere” is a collision of the Bloch drawing, the state space, and the metric of the room. Useful as a nickname. Useless as a derivation.

What the joke is trying to say

Stripped of the planet, the worry is ordinary and worth keeping:

- A state space can be enlarged — more modes, more particles, a bigger cutoff — without the enlargement being a physical stretching of the sky.
- A geometric picture (curvature, a defect, a hole) can be a metaphor for missing structure: a term the equation does not have, a boundary condition nobody wrote down, a sector the model cannot represent.

- If the missing structure grows with the size of the model, then “more complete” and “more wrong in a new place” can happen together. That is the only sense in which a bigger fabric makes a bigger hole.

A planet falls out only if the proposal is secretly claiming that physical geometry and state-space geometry are the same geometry. The front page does not claim that. It asks whether spatial geometry might mediate quantum interactions. Those are different sentences. The second one can be probed. The first one is a cartoon.

Established mechanics, so the modification has somewhere to stand

Ordinary quantum mechanics already has a clean split. The Hamiltonian generates time evolution, $U(t) = \exp(-iHt/\hbar)$, on a Hilbert space fixed by the system you chose. For a single particle in a potential the Hamiltonian is an operator, $H = p^2/2m + V(r)$, acting on wavefunctions. The wavefunction Ψ is what the operator acts on. It is not a term inside H .

The front-page line $H = H_0 + (\hbar^2/2m) \nabla^2 \Psi + V(r,t)$ puts an operator, a wavefunction, and a potential in one sum. That is a hole you can see without leaving the homepage. A cleaner split, still only a starting point, is: pick the space, write the operator, then let it act.

$$H = H_0 + T[g] + V, \quad T[g] \Psi = -(\hbar^2/2m) (1/\sqrt{|g|}) \partial_\mu (\sqrt{|g|} g^{\mu\nu} \partial_\nu \Psi).$$

The sign and the factor of $\sqrt{|g|}$ are the Laplace–Beltrami operator, not a new postulate. If the metric g is the ordinary flat metric, this reduces to the textbook kinetic term. If g is promoted to something dynamical, or to something that depends on the state, that promotion is the proposal, and it has to be written as an assumption, not smuggled into the notation.

Assumptions a visitor is allowed to remove

1. The background metric is prescribed, not back-reacting. Remove this and you owe a rule for how g changes.
2. The Hilbert space is the space of square-integrable functions on that background. Remove this and you owe a different state space.
3. Measurement is still the Born rule on that space. The site asks whether geometry could influence measurement. That question is open only after someone states which part of the measurement postulate moved.
4. “Spatial” means the metric of physical space, not the Fubini–Study metric on projective Hilbert space. Conflating those two is the Hilbert-sphere joke in formal clothes.

Any of those four can be false. Saying so is not a defense of the proposal. It is the proposal’s only honest shape.

Questions that would actually dismantle it

- What observable changes if $T[g]$ is used, that does not already change in ordinary quantum mechanics on a curved background? Curved-background quantum mechanics already exists. A new name is not a new prediction.
- Where is the smallest experiment: one qubit, one interferometer, one number that differs? If the answer needs a planet, the answer is too big.
- Which limit recovers textbook quantum mechanics exactly, and which limit is forbidden?
- Is the Bohr Flop in C minor evidence? No. It is a sonification of a driven qubit and an oscillator superposition. Personality is allowed. Personality is not a dataset.

A useful reply is a broken assumption, a missing reference, or a number. “This is speculative fiction” is also a useful reply, and the site already asks for it.

Who threw this down

This note is a write-up of a voice session on 6 October 2026. The poke-holes line came back from a read of quantumspatial.lol. The planet, the expanding fabric, and the refusal to sound like a podium came from the person who owns the site. The separation into Hilbert space, Bloch sphere, and physical metric, and the repair of the homepage operator, were written here so the joke would have a floor.

It is not a product of a named institute. It is not a claim that a model has been derived. If a later page cites this file, cite it as a site note: conversation, then definitions, then the holes.

One paragraph, if the homepage only has room for the door

Poking holes is the point. “Hilbert sphere” is already a mixed picture: a Hilbert space is the state space, the Bloch sphere is a drawing of one qubit, and physical space is where a planet could actually be. Stretching one of those does not automatically tear the others. The homepage sum that puts $\nabla^2\Psi$ inside H is a hole you can see without a metaphor. A cleaner start is an operator $T[g]$ that falls back to the textbook kinetic term when g is flat, and is labeled as a proposal the moment g is allowed to depend on the state. Send the broken assumption. The planet can stay in the joke.

Suggested placement: under Books and papers, or as its own short section between Questions to Explore and We Want Your Thoughts. Title on the site can stay “Poking Holes in the Hilbert Sphere.” The subtitle should keep the warning: a picture, not a derivation.